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Modern distribution system expansion planning considering new market designs: Review and future directions

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ABSTRACT

The distribution system is undergoing profound transformations with decarbonization efforts through the rapid integration of renewable energy and the consequent transition of the distribution network from passive to active. Nonetheless, environmental and economic urgency requires redesigning the existing distribution system. New operational standards and technologies such as electric vehicles, demand response, energy storage systems, energy hubs, microgrids, and transactive energy markets have been developed. Distribution system expansion planning plays a crucial role in supporting this transition process. This work presents a review of planning proposals published in the last decade considering flexibility, local markets, and microgrid concepts. The main aspects of these works are described in terms of solution methods, expansion alternatives, planning periods, uncertainties, and contributions, among others. Furthermore, planning proposals are analyzed and classified taking into account new market designs. In addition, future trends are identified. Finally, this review can be used as a tool and guide for different stakeholders on current practices and new trends regarding the distribution system expansion planning, serving as a foundation for future investigations.

Nomenclature

Abbreviations	
CBs	Capacitor banks
CHP	Combined heat and power
DERs	Distributed energy resources
DG	Distributed generation
DISCOs	Distribution companies
DR	Demand response
DSEP	Distribution system expansion planning
DSO	Distribution system operator
EH	Energy hub
ESSs	Energy storage systems
EVCSs	Electric vehicles charging stations
EVs	Electric vehicles
GA	Genetic algorithm
GENCOs	Generating companies
GPL	Gridable parking lots
ISO	Independent system operator
LEMs	Local energy markets
MCS	Monte Carlo Simulation
MILP	Mixed-integer linear programming
MINLP	Mixed-integer non-linear programming

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Abbreviations	
MISOCP	Mixed-integer second-order cone programming
PSO	Particle Swarm Optimization
PV	Photovoltaic
REM	Retail energy market
TS	Tabu Search
VRs	Voltage regulators
WEM	Wholesale energy market
WT	Wind turbine
Indexes	
a	Index of conductor types
i/ij	Index of nodes/circuits
l	Index of loads
p	Index of planning period
Parameters	
Δ_l	Number of hours at load level l
λ	Number of years in each period
τ	Discount rate
C^{CB}	Cost for capacitor bank installation
$C^{L}_{ij,a}$	Cost for the installation of circuit ij using conductor type a
C^{LOSS}	Cost related to energy losses
C^{S}_{i}	Cost for substation installation at node i

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(*continued*)

Abbreviations	
C^{VR}	Cost for voltage regulator installation
l_{ij}	Length of circuit *ij*
R_a	Conductor resistance
Variables	
$I_{ij,a,p}$	Current through circuit *ij* for conductor *a* and period *p*
ICB_p/ IVR_p	Investments in capacitor banks/voltage regulators
IS_p/ IC_p	Investments in substations/circuits
$N^{CB}_{i,p}$	Number of capacitor bank modules installed at node *i*
TC	Total cost
$x^{L}_{ij,a,p}$	Construction of circuit *ij* using conductor type *a* in period *p*
$x^{S}_{i,p}$	Substation reinforcement/construction at node *i* and period *p*
$x^{VR}_{i,p}$	Voltage regulator allocation at node *i* and period *p*

1. Introduction

In the last decade, the global dependence on electricity and concerns about environmental issues have driven power and energy systems into a new era, encouraging the rapid integration of renewables. In addition, the Russia-Ukraine war resulted in an energy crisis (mainly in countries on the European continent), creating a new urgency to accelerate the clean energy transitions and highlighting the crucial role of renewable generation [1]. In this context, new world policies have contributed to the growth of such technologies, with governments setting ambitious targets to reduce greenhouse gas emissions. For example, the European Union has adopted a target of reducing at least 55 % of net emissions by 2030, relative to the 1990 level [2]. These policies bring a new paradigm to distribution systems demanding profound transformations.

In this transition process, new technologies, operational patterns, and players have emerged (e.g., flexibility resources, flexibility services, transactive energy markets, and prosumers). The flexible resources can include technologies such as distributed generation (DG), electric vehicles (EVs), energy storage systems (ESSs), and demand response (DR) [3]. Such resources can provide flexibility services in distribution systems where end-users have been seen as passive participants so far. However, with the increasing integration of distributed energy resources (DERs) in the distribution system and the transition of the distribution network from passive to active, the end-user gains a new role, being able to provide flexibility services for the distribution system operator (DSO) as well as providing security for their supply of energy (e.g., using photovoltaic (PV) units jointly with ESSs) [4]. In fact, these changes open a new era for the power system, bringing many challenges to the operation and planning of distribution systems.

In such a context, new market designs have been seen as one solution to deal with the challenges posed by the modern power system. The different proposed frameworks come as a tool to support the integration of flexibility resources meeting different stakeholder needs (e.g., DER investors, DSO, and consumers) [5]. Local energy markets (LEMs) emerge in this situation to provide more efficient use of DERs within a specific area, allowing small consumers to have a more active role in the energy market [6]. The presence of these new LEM mechanisms can have a significant impact on the distribution expansion planning (DSEP). Therefore, a new way of planning distribution systems must capture the characteristics of these emerging markets to encourage the development of a sustainable power system. Despite the vast literature studying the impact of LEMs into the operation of distribution systems, the issues that might arise related to the DSEP have been little explored in current research. DSEP has played a crucial role in meeting the new configuration design of power systems in recent years, ensuring adequate operation performance in terms of reliability, safety, power quality, environmental goals, and sustainability. The solutions to DSEP problems determine what investments are needed in the distribution system to meet the growing demand and fulfill a set of constraints (e.g., power balance, voltage and current limits, radiality, budget limit) at the lowest possible cost [7].

The traditional optimization planning problem addresses the reinforcement or installation of existing and new substations and circuits [8]. This problem has become more complex with the penetration of DERs and the decentralization of power systems. Fig. 1 illustrates the DSEP problem framework showing some examples of traditional and modern investment possibilities for the distribution systems. Conventional planning usually invests in constructing new substations/circuits and reinforcing existing substations/circuits to expand the distribution network. Furthermore, capacitor banks (CBs) and voltage regulators (VRs) are expansion alternatives usually considered in short-term planning proposals [9]. Also, with the increasing adoption of DERs, other decision variables have been added to the planning problem, such as the allocation of DG units, EVCSs, and ESSs.

Several review articles have addressed the DSEP problem considering different aspects and points of view. For instance, the state-of-the-art for the optimal planning of DERs is presented in Ref. [10], emphasizing multi-objective approaches in terms of solution techniques, decision-making methods, multi-objective performance indexes, and DER technologies; the authors highlighted the use of ε-constrained methods and multi-objective evolutionary algorithms (e.g., NSGA-II), which are techniques commonly used to solve multi-objective problems. In other ways, the review of DSEP approaches presented in Ref. [11] and published in 2015 discusses the general aspects of the problem, establishing a critical analysis of the planning proposals. A following work [12], published in 2018, reviewed the planning problem of renewable DG, highlighting analytical solution methods grouped into six categories according to different characteristics (e.g., exact loss equation, loss sensitivity factor, etc.). Also, the authors compared analytical techniques in terms of objective function, number and type of DG units, and decision variables, followed by a series of observations and recommendations for the planning of renewable generation in distribution networks. One of the relevant points mentioned in that work is the integration of ESSs with renewable resources, a configuration that helps to reduce the impact of uncertainties of renewables and brings more reliability to the distribution system. More recently, in 2019, the survey article [13] presents state-of-the-art methods for modeling uncertainties applied to the planning of active distribution networks. These methods are discussed in depth and divided into six groups: probabilistic techniques, stochastic optimization, robust optimization, possibilistic techniques, hybrid probabilistic–possibilistic techniques, and information gap decision theory. In addition, it provides a critical discussion and a comparative analysis of uncertainty modeling methods.

Finally, a study published in 2020 [14], presents an overview of the proposals for the DSEP problem, describing the main criteria for optimization, i.e., decision variables, objective function, and constraints. Furthermore, the authors identified and discussed the most used methods to solve the problem (mathematical and heuristic techniques). Other issues briefly mentioned by the authors include uncertainty modeling and multi-objective optimization methods. To the best of our knowledge, this is the only review in which the concept of LEMs in the DSEP problem has been highlighted as a key point for future research.

In the past few years, research in the DSEP area has included several simultaneous participants (e.g., prosumers, DSO, distribution companies (DISCOs) in the investment decision process. However, the review articles cited previously mainly focus on centralized planning strategies, i.e., proposals that determine expansion planning decisions based on a single agent perspective. Most works found in specialized literature adopt centralized planning. Furthermore, the consideration and impact of local markets in the planning problem is also recent, and no review article has yet covered this aspect. Therefore, this survey fills these gaps in the specialized literature, identifying and deeply discussing proposals addressing the DSEP problem considering new market designs. Moreover, the state-of-the-art of the DSEP problem is reviewed considering the flexibility resources [15,16]. From this investigation, this work seeks to answer the following research question: Are the current proposals of

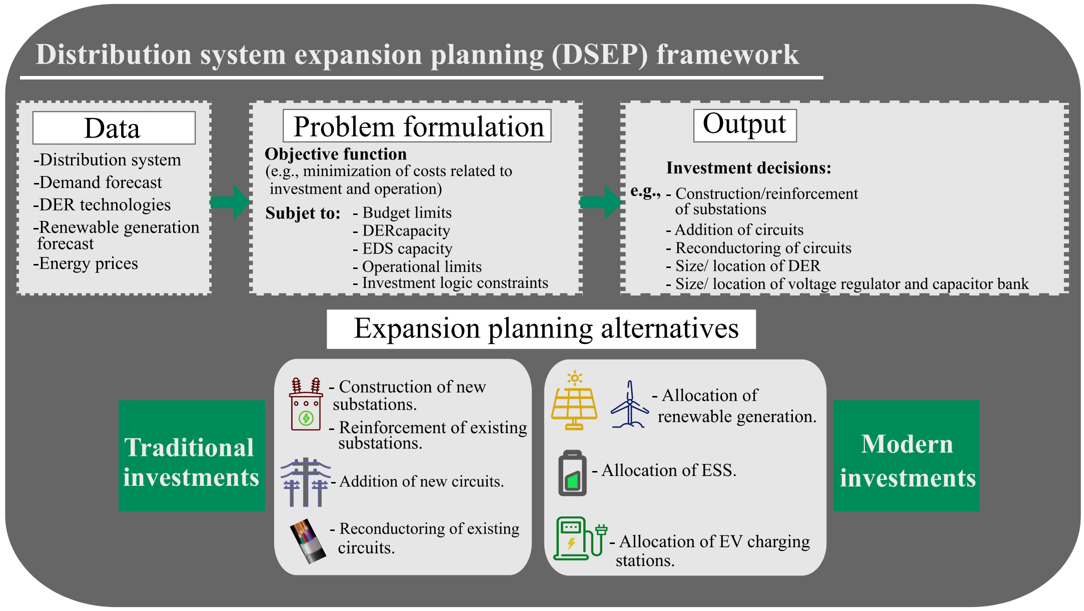


Fig. 1. Distribution system expansion planning (DSEP) framework.

distribution system expansion planning in line with the transformations that have been taking place in power systems? Thus, we investigated recent planning proposals to answer the guiding question of our research.

The main contribution of this work lies in offering a contemporary view of the DSEP problem. Unlike conventional approaches focused on system expansion through adding assets, this research also explores strategies that avoid network infrastructure upgrades. Here, it investigates innovative paths such as energy management and new market designs, shedding light on non-network solutions. More importantly, this research provides valuable insights into the future trajectory of the DSEP problem, identifying emerging trends and potential directions for further exploration. By providing insight into these alternative approaches, the work enriches comprehension of DSEP, offering valuable guidance for navigating the evolving energy landscape.

A summary of the main contributions of this work is presented as follows.

i. A review of planning strategies that avoid upgrading grid infrastructure reveals that these approaches exploit existing resources in the power system, intelligently managing resources such as flexible services through DERs, smart energy management, and participation in energy markets to optimize energy usage and balance supply and demand without the need for extensive physical expansion.
ii. A critical analysis of proposals published in the last decade addressing the DSEP problem considering flexibility resources, categorizing the works in terms of models, methods, types of investments, and modeling of uncertainties.
iii. The first review in the specialized literature that relates the DSEP problem to energy markets, providing an in-depth discussion about the state-of-the-art and identifying the main aspects of the reviewed works.
iv Identification of the existing gaps in the DSEP problem and discussion of paths for future research.

The remainder of this work is structured as follows. Section 2 describes the main aspects of the planning problem, covering variables, planning period, solution methods, uncertainties, and planning strategies. Sections 3, 4, and 5 analyze different planning alternatives: traditional expansion alternatives (Section 3), modern expansion alternatives and flexible resources (Section 4), and non-network solutions (Section 5). Section 6 presents a detailed review of the planning proposals considering energy trading, highlighting the main market characteristics and the DSO responsibilities. Section 7 provides an in-depth discussion and analysis of planning proposals in the current literature, highlighting their strengths and weaknesses. Finally, Section 8 outlines trends for future research, and Section 9 draws conclusions.

2. Main aspects of the problem

The traditional DSEP problem aims to determine, in the most economical way, the investments needed in the distribution network to meet the growing demand, maintain the safe operation of the system, and guarantee the quality of supplying electricity to consumers. The DSEP problem correspond to a mixed-integer non-linear programming (MINLP) model since it includes continuous and integer variables under a set of non-linear equations modeling the distribution network operation. Solving this problem is a complex task due to its combinatorial nature and the significant amount of continuous and integer decision variables [11].

This section presents a summary of the main characteristics of the planning problem such as the planning period, variables, methods most used in solving the problem, uncertainties, and planning strategies.

2.1. Planning period

According to the optimization model, the problem approach can be single-period or multi-period planning. In the former, the planning is carried out in only one period; thus, demand forecast for the end of the planning horizon is used in the proposed planning. Multi-period planning, is divided into several periods in which expansion actions are carried out at different times along the planning horizon according to the corresponding demand forecast [7,17]. Multi-period planning is a more realistic approach since investments are made in different periods according to the corresponding needs [18]. However, many authors have adopted static (single-period) planning to reduce the complexity of the problem [19–21].

2.2. Variables

Integer/binary and continuous variables constitute the DSEP problem. The binary/integer variables are decision variables of the problem, and they generally define the location, size, type, and planning period of the asset installed in the distribution network. The continuous variables are defined after the decision variables of the problem, e.g., power supplied by the substation to the distribution network, voltage magnitude at a node, and active/reactive power flow through a circuit, among others. Therefore, the planning problem has many variables, and as more decision variables are incorporated, the complexity of the problem increases.

2.3. Solution methods for the DSEP problem

Mathematical programming and heuristic/metaheuristic methods are among the most used methods to solve the planning problem. This section briefly describes the advantages and some characteristics of these methods.

- Mathematical programming: Mathematical programming is one of the most adopted methods to solve the DSEP problem. It has the advantage of guaranteeing the optimality of the solution found. Furthermore, mathematical programming formulations are simpler to implement than heuristics/metaheuristics methods and can be solved via commercial solvers, e.g., CPLEX [22] or GUROBI [23]. In this context, DSEP is often modeled as a mixed-integer linear programming (MILP) problem [24–30]. Another widely used formulation for the problem is the mixed-integer second-order cone programming (MISOCP) model [31–33]. The DSEP problem contains integer and binary variables, which makes the problem complex. The Bender's decomposition method has been proposed to deal with this problem [32]; this technique decomposes the main problem into subproblems, reducing the computational time. Another method used in these problems is the column-and-constraint generation algorithm that simplify and solve the DSEP problem in parts [34]. Moreover, dynamic programming has been used to solve the DSEP problem in a multi-objective approach considering reliability [35]. However, such a method has a high computational cost to solve large-scale planning problems [14].
- Heuristic and metaheuristic methods: Exact solution techniques present difficulties or may not be able to solve large-scale planning problems. In that context, heuristic techniques are advantageous. Heuristic techniques have been widely used in multi-objective approaches for the minimization of costs and maximization of reliability, e.g., Tabu Search (TS) [36]and Particle Swarm Optimization (PSO) [19]. Such methods have been frequently adopted due to their versatility in modeling different parts of the DSEP problem. For instance, the branch-exchange heuristic has been directly applied to the long-term planning problem [18] and within the implementation of Genetic (GA) and Evolutionary algorithms [20,37]. Other methods applied to solve DSEP problem are the Imperialist Competitive Algorithm [38,39], Heuristic Moment Matching [40], Artificial Bee Colony [41], Cuckoo Optimization Algorithm [42], and Parallel Bird Swarm Algorithm [43]. However, unlike exact solution techniques, these methods do not guarantee the optimality of the solutions found.

2.4. Uncertainties

The uncertainties are inherent to the DSEP problem. In the specialized literature, the most addressed uncertainties are hourly variation in the level of demand, electricity price, solar irradiance, wind speed, the output power of renewable generators (e.g., PV units, wind turbine (WT) units), and EV demand. Such uncertainties are short-term, while long-term uncertainties, e.g., uncertainties related to demand growth and DG penetration over the planning periods, have not been properly explored. Since, long-term uncertainties significantly influence planning decisions, addressing them can improve the quality of the solution found for the DSEP problem.

Stochastic programming has been widely used to deal with sets of uncertain parameters, and often planning problems have been formulated as a two-stage stochastic programming model [8,44]. In this approach, investment decisions (e.g., construction of substations/circuits, allocation of DG units) are determined before the realization of the uncertainties. For this reason, these decisions are called "here and now." The second stage defines the system operation that occurs after realizing uncertainties; examples of second-stage variables are power injected by generators, power supplied by the grid, and voltage at nodes. These second-stage decisions are commonly called "wait-and-see". In this model, uncertain parameters are usually represented by discrete scenarios, and each scenario has a probability of occurrence. Thus, stochastic programming models define their decisions based on an objective function representing an expected value (e.g., minimizing expected cost, maximizing expected profit). One issue with this approach is that the decisions are risk-neutral, not necessarily catering for worst-case scenarios with low probability but a high impact on the quality of the solutions. The stochastic programming model can be extended to get around this problem by incorporating risk measures such as CVaR, which have been applied in risk-averse planning problems [42]. One of the main disadvantages of stochastic programming is its high computational cost that increases linearly with the sample size. Therefore, such an approach may not be scalable for problems with a large set of scenarios.

Chance-constraint method has been used in problems under uncertainty to provide a certain flexibility, allowing the constraints to be fulfilled within a confidence level [45]. Unlike the chance-constraint method, robust optimization [9] guarantees that the solution found unaffected to all data uncertainties within a defined set, i.e., the solution is feasible for all scenarios, including the worst-case scenario. Nonetheless, the disadvantage of this method is that it produces conservative solutions.

The previously mentioned optimization methods (stochastic programming, probabilistic constraints, robust optimization) are applied in formulating problems under uncertainties. However, before applying these methods, it is necessary to model the uncertainties in the planning model. Techniques used to represent uncertainties include.

- Historical data: Analyzing historical data offers valuable perspectives on previous trends, fluctuations, and patterns, which can be crucial for modeling uncertainties. Conversely, the available historical data may be limited in terms of sample size, especially for rare events [46].
- *K*-means clustering: This approach in the DSEP problem is usually applied to historical data to group the data and reduce scenarios. Since it is dependent on the number of clusters (k), choosing an inappropriate *k* value can result in suboptimal clustering [47].
- Probabilistic modeling: This method recognizes the inherent variability in the parameters or variables of interest and integrates probability distributions to measure and analyze these uncertainties. Such a method might encounter challenges in accurately representing extreme events or tail risks, especially when there is limited historical data on such occurrences [44].
- Monte Carlo Simulation (MCS): MCS has been used to model the probability of uncertain parameters in a process that cannot easily be predicted due to the presence of random variables. This method might not be the most suitable for modeling rare events or extreme outcomes, as generating an adequate number of samples for such infrequent occurrences can be challenging [48].
- Autoregressive Integrated Moving Average (ARIMA) model: ARIMA models are widely embraced and frequently employed as a statistical technique for forecasting time series data. These models are typically

more effective in forecasting for the short to medium term. Their accuracy tends to diminish when applied to longer time horizons, leading to less reliable forecasts [49].

- Fuzzy C-means: In contrast to conventional clustering approaches, fuzzy C-means allocates membership degrees to data points, enabling them to simultaneously belong to multiple clusters. Such a method is sensitive to the initial values of cluster centers and membership degrees. Different initializations can lead to different results, making the algorithm less robust [41]. Finally, Table 1 categorizes different aspects of modeling uncertainties related to uncertain parameters, optimization methods under uncertainty, and methods to model the uncertain parameters.

2.5. Planning strategy

Most planning proposals have been developed under a centralized planning strategy. In such an approach, planning decisions are made simultaneously based on the interests of a single agent. Furthermore, in some proposals, the DSO or DISCOS are responsible for planning the distribution network and DERs, being the DERs owners in some cases. However, this practice adopted in many planning proposals may be inadequate since the existing legislation does not allow the DSO to own and operate the DERs. Only in some specific cases, the DSO can own DERs, e.g., when no other agent has expressed an interest or when the installation of storage systems is necessary for the DSO to fulfil its attributions; even in these cases, the approval of the national regulator is required [59]. Therefore, planning proposals must address the interests of different actors (e.g., DER investors, prosumers, and EVCS owners), delimiting the attributions of each one. The following are some examples of planning proposals that consider the interests of various agents: (1) The authors in Ref. [15] proposed an incentive-based model that encourages investors to install DG units in specific nodes in a way that benefits DG owners and the DISCO; (2) Some proposals to address different interests in the distribution network and DERs planning formulate the problem as a bilevel programming model [60].

In the specialized literature, the planning of energy systems is commonly carried out independently. However, recent research has developed proposals for the integrated planning of multiple energy systems such as electricity, heating systems, hydrogen, and natural gas [61,62]. The interest in this strategy is due to the emergence and growth of new technologies, e.g., CHP, electric heat pumps, EVs, and hydrogen fuel vehicles. In this context, some investigations highlight that the planning of energy systems should encompass other sectors, not just electricity, for more effective planning. One concept explored within this planning approach is the Energy Hub (EH). Such a concept is defined in Ref. [63] as "*The place where the production, conversion, storage and consumption of different energy carriers takes place, is a promising option for integrated management of multi-energy systems*". Such a concept has been explored to deal with the challenges of the new era of power systems, promoting the integration of multiple energy resources collaboratively [64,65].

3. Traditional expansion alternatives

Traditional expansion alternatives mainly include investments associated with the construction and reinforcement of substations [15], installation of new circuits [66], and reconductoring of existing circuits [16]. Furthermore, investments in CBs and VRs [24] are also part of the traditional planning, although mainly considered in the short-term horizon [9]. Traditional DSEP is often formulated as the minimization of total costs (TC) corresponding to the sum of investments (fixed) and operation (variable) costs [20,67], as shown in (1). The decision variables of the problem are related to the investments made, where IS are the investments in substations (2), IC are the investments in circuits (3), ICB are the investments in capacitor banks (4), and IVR are the investments in voltage regulators (5). The mathematical model has binary and integer variables regarding the installation of assets in the distribution network, e.g., $x^{\mathrm{L}}_{ij,a,p}$ represents the construction of circuit ij using conductor type a in period p; $x^{\mathrm{S}}_{i,p}$ represents the substation reinforcement/construction at node I and period p; $x^{\mathrm{VR}}_{i,p}$ represents the voltage regulator allocation at node I and period p; and the integer variable $N^{\mathrm{CB}}_{i,p}$ indicates the number of capacitor bank modules installed at node i. Equation (6) presents the operational costs (variable costs) related to energy losses, which are the losses most commonly addressed in the traditional DSEP. C^{LOSS} is the cost parameter related to energy losses; Δ_l is the number of hours at load level l; R_a represents the conductor resistance; and $I_{ij,a,p}$ represents the current through circuit ij for conductor a and period p. Furthermore, the function $f(\tau,\lambda) = \left(1 - (1+\tau)^{-\lambda}\right)/\tau$ allows the calculation of the present value of a given annualized cost in terms of the number of periods λ and the discount rate τ.

$$\min TC = \sum_{p} \left(\textit{Fixed costs}_p + \textit{Variable costs}_p\right)\,(1+\tau)^{-(p-1)\lambda} \tag{1}$$

$$IS_p = \sum_{i} C^{\mathrm{S}}_i x^{\mathrm{S}}_{i,p} \quad ;\forall p \tag{2}$$

$$IC_p = \sum_{ij}\sum_{a} C^{\mathrm{L}}_{ij,a} l_{ij} x^{\mathrm{L}}_{ij,a,p} ;\forall p \tag{3}$$

Table 1
Main aspects of planning proposals under uncertainties.

Ref.	Uncertain parameters					Optimization methods				Methods to model the uncertain parameters				
	Demand	EV demand	Solar radiation	Wind speed	RDG	SP	RBM	CCM	RO	K-means	HMM	ARIMA	Fuzzy C-means	PM/or MCS
[8,16]	✓	✓	✓	✓		✓				✓				
[9,50]	✓				✓				✓					
[40,51]	✓		✓	✓		✓					✓			
[41]					✓								✓	
[44,52]	✓		✓	✓		✓								
[53]	✓	✓	✓	✓		✓	✓			✓				
[54]	✓		✓	✓				✓						✓
[55]	✓				✓	✓				✓				
[56]	✓				✓	✓								✓
[48,57]	✓				✓									✓
[58]	✓				✓				✓	✓				
[49]	✓			✓		✓						✓		

RDG: renewable DG units; SP: Stochastic programming; RBM: Risk-based model; CCM: Chance-constraint method; RO: Robust optimization.
PM/or MCS: Probabilistic model and/or Monte Carlo simulation.

$$ICB_p = \sum_i C^{\text{CB}} N_{i,p}^{\text{CB}} \quad ; \forall p \tag{4}$$

$$IVR_p = \sum_i C^{\text{VR}} x_{i,p}^{\text{VR}} \quad ; \forall p \tag{5}$$

$$O_p^{\text{LOSS}} = f(\tau,\lambda) \sum_l \sum_{ij} \sum_a C^{\text{LOSS}} \Delta_l R_a I_{ij,a,p}^2 ; \forall p \tag{6}$$

4. Modern expansion alternatives and flexible resources

Modern expansion alternatives and flexible resources focus on incorporating advanced technologies, improving sustainability, and addressing the evolving energy landscape. In this section, some contemporary expansion alternatives explored in the DSEP problem are presented.

4.1. Distributed energy resource integration

Investments in DERs are explored as expansion alternatives in the contemporary landscape. PV, WT, ESS, and non-renewable DG units are the most used modern expansion alternatives [44,47]. The allocation and size of these technologies are incorporated as decision variables in the planning proposals. In these proposals, the planning problem aims to meet demand through reinforcement, i.e., increasing the capacity of the distribution system by reinforcing or adding assets to the network [53]. In the DSEP problem, DERs have commonly been considered as an alternative for reinforcing the distribution network [68,69], and their potential as a flexibility resource has been little explored in the context of planning.

Studies in the DSEP problem area that address DERs present some key characteristics in the proposed models: (1) Location, size, and time: Proposed models often determine the optimal siting and sizing of DERs. Furthermore, some proposals consider multi-period models that define the ideal period to invest in DERs [16]. (2) Modeling renewable generation output: Planning models usually address the variability and intermittence of renewable generation by incorporating the uncertainties associated with these resources into the model [44,70]. (3) Optimization priority: Optimization models are often used to find the most cost-effective solutions for DER integration [71], considering a series of technical and operational constraints.

The proper integration of DERs brings flexibility to the distribution system, providing many benefits such reduction of network congestion, support in peak periods, and voltage control, improving the system's reliability. In addition, DERs can contribute to the decarbonization of the energy and transport sectors, mainly through renewable generation and zero-emission vehicles. The renewable generation technologies most addressed in the specialized literature are PV units and WT units [72,50], which are the fastest-growing renewable technologies worldwide [73]. The growth trend of these technologies should continue in the coming years, encouraged by the environmental goals and the reduction of investment costs associated with these technologies [74]. Other types of DG considered in the planning proposals are micro-hydro turbines [72], fuel cells [75], biomass DG units [16], and dispatchable DG units [8,76].

A disadvantage of renewable generation is the uncertain behavior related to its energy production, e.g., solar and wind generation are highly dependent on weather conditions (i.e., solar irradiation and wind speed). In this context, energy storage technologies can be used to overcome this issue and reduce the impact of uncertainties associated with renewable generation. Furthermore, ESSs can support the grid by providing additional capacity when needed, playing a key role in microgrids [50,54,77]. Thus, past studies have considered ESSs as an alternative investment in planning problems [42,55,78]. Within the concept of DERs, DR is another tool that contributes to the more efficient use of electricity, e.g., by supporting the integration of renewable generation and bringing flexibility to the distribution system. The International Energy Agency (IEA) defines DR as follows: "*Demand response involves shifting or shedding electricity demand to provide flexibility in wholesale and ancillary power markets, helping to balance the grid. It is based on two main mechanisms: price-based programmes (or implicit demand response), which use price signals and tariffs to incentivise consumers to shift consumption, and incentive-based programmes (or explicit demand response), which monetise flexibility through direct payments to consumers who shift demand in a demand-side response programme*" [79]. The DR concept allows consumers to take a more active role in power systems, allowing changes in the electricity consumption patterns of end-users in response to changes in electricity prices over time or incentives offered to reduce electricity use during periods of high energy prices or compromised system reliability. Planning proposals with investments in renewable generation [78,80] show the impact of considering DR to time-varying prices, providing a reduction or postponement in investments in distribution expansion.

DERs can significantly influence DSEP problem decisions, impacting the problem in the following aspects.

- Peak demand reduction: Pairing ESSs with renewable generation enables effective peak shaving by storing surplus energy during periods of low demand and delivering it when demand is at its highest. This can mitigate the necessity for costly infrastructure upgrades to handle peak loads [81]. Also, DR initiatives aid in peak demand management by offering incentives for consumers to curtail their electricity consumption during periods of increased demand. This has the potential to postpone the requirement for infrastructure expansion in the DSEP problem [78].
- Quality improvement: DERs can play a key role in voltage regulation and enhancing power quality. This capability can diminish the requirement for equipment upgrades, ensuring voltage levels within statutory limits [82].
- Decentralized generation: DERs introduce a more decentralized approach to power generation, fostering a more flexible and adaptable distribution system in contrast to a centralized system with inflexible expansion plans [53,83].
- Distribution system modernization: The deployment of DERs frequently requires efforts to modernize the grid. This involves incorporating smart grid technologies, advanced communication systems, and digital control systems to effectively manage and optimize the interaction between the distribution system and DER [14].
- Charging infrastructure planning: The DSEP problem needs to incorporate the development and deployment of charging infrastructure. This includes the strategic placement of charging stations to support the growing number of EVs and to avoid overloading specific grid segments [45].
- Regulations: The impact of DERs on the DSEP problem is significantly shaped by regulations and policies. Incentives, tariffs, and regulatory frameworks can notably influence on the deployment of DER and its integration into the existing infrastructure [84].

Finally, Fig. 2 highlights the main technologies and concepts related to DERs considered by the specialized literature in the planning problem.

4.2. Electric vehicle charging stations

The integration of EVCSs in the DSEP problem is crucial to support the growing adoption of EVs. In the last decade, efforts have been directed to address the simultaneous planning of EVCSs and distribution systems, highlighting the impact of the EVs charging demand in investment decisions in distribution systems [45,85]. According to the literature review, some common characteristics are found in the EVCS planning proposals.

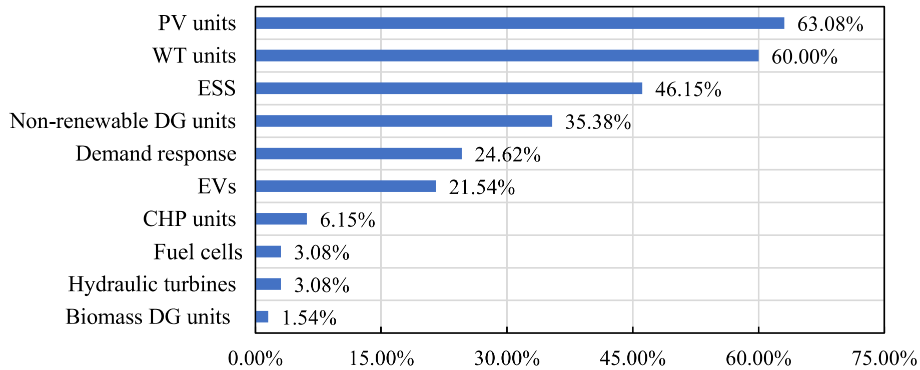


Fig. 2. Percentage frequency of DER technologies in the reviewed works.

- Size and location: EVCS planning defines the capacity and ideal location of stations to meet the EVs demand at the lowest possible cost while meeting a series of physical, technical, and operational constraints. Such planning determines the power and quantity of chargers installed in the stations. In addition, the integrated planning of EVCSs and distribution systems also addresses the installation of new substations/circuits and reinforcement in existing substations/circuits [86,87]. Some proposals investigate the design of DG units, especially renewable technologies [88,89]. Integer and binary variables represent the investment decision variables of this problem, making the DSEP even more complex.

- Uncertainties in the EV charging demand: Uncertainty related to EVs is often addressed in the EVCS planning problem. Several methods and models have been proposed to determine the EV demand. Due to the lack of real data on the behavior of EV users, many studies have used travel data surveys from conventional vehicles [85].
- Centralized planning: The centralized planning strategy has often been adopted in the planning of EVCSs, DG units, and distribution systems [16]. In such a strategy, the individual interests of the owners of EVCSs and DG units are not considered, and the DISCO or DSO defines the investments in distribution systems, EVCSs, and DG units. This strategy is not the most realistic and suitable, but it is adopted to simplify the DSEP.
- Integration with renewable generation: The integration of EVCSs and renewable generation has been widely considered in specialized literature [87,90], to promote sustainability. This may involve coupling charging stations PV/WT units, reducing the overall environmental impact.

4.3. Microgrids

Microgrids have emerged as a tool of flexibility in modern power systems, this concept collaborates with the development and integration of DERs. Microgrids are relatively small energy systems composed of DERs connected to nearby users. Microgrids are decentralized systems, generating, storing, and distributing energy locally. Also, a microgrid can be operated independently (island mode) or connected to the primary grid. This technology offers an alternative to the external distribution network and contributes to adopting DERs in the power system [91–93]. Microgrids can be designed to supply heating and electrical loads, as in Refs. [72,75]. Moreover, some proposals incorporate cooling loads [77]. The following technologies can be used to meet these different loads:

- Combined heat and power (CHP): The CHP is a type of generator that can simultaneously produce electricity and thermal energy (heating or cooling). Such technology has been used in microgrid planning [42,94].
- Gas-fired boilers: Gas-fired boilers are devices used in heating systems; such equipment uses gas as a form of fuel to heat water. The gas boiler heats and stores the water to meet the heating demand [77,94].
- Thermal energy storage (TES): TES is a technology that transfers heat to a storage medium to be used at some moment in heating and cooling applications or power generation [77].

The growing integration of microgrids into power systems highlights the need for strategic planning of these emerging structures in the DSEP problem. In recent years, there has been a notable surge in research attention towards the planning of distribution systems with a focus on microgrids [94,95]. Such approaches have proposed models for planning grid-connected microgrids, which allow energy exchange between the main grid and the microgrids according to their needs. Furthermore, these planning strategies require the involvement and collaboration of different stakeholders, e.g., DSO and microgrid investors [96,97].

Table 2 highlights some aspects related to the classification of microgrids, such as types of generation, storage systems, grid connection, and other types of energy supply besides electricity (e.g., thermal energy). PV, WT, and dispatchable DG units have been the most widely considered technologies for power generation (see Table 1). Other technologies, such as electric vehicle charging stations (EVCSs), were considered in planning of microgrids as proposed in Refs. [98,99]. The microgrid planning problem commonly defines the size and location of DG units; scheduling of generation and storage to minimize operating costs; and the location and time of installation of microgrids. Furthermore, in some cases, switch allocation is used to establish the electrical boundaries of microgrids [42,77]. Finally, a study of computational methods applied to microgrid design has been explored by Ref. [100],

Table 2
Generation/storage technologies, grid connection, and other aspects of microgrids.

Classification		Reference
Type of generation	PV units	[42,72,50,75,54,77,98]
	WT units	[72,50,75,54,77]
	Dispatchable DG units	[72,50,75,54,98]
	Fuel cell	[75,77]
	CHP units	[42,77]
	Micro-Hydro	[72]
Storage systems	ESSs	[72,50,75,54,77,98]
	TES	[72,77]
Other technologies	EVCSs	[98]
	Switching device	[42,77]
Network connection	External network connection	[72,50,77]
	Islanded mode	[54,98,99]
District energy networks	Thermal energy supply (heating and/or cooling)	[72,75,77]

while the authors in Ref. [101] reviewed important aspects of microgrids, such as technologies, challenges, and future perspectives.

5. Non-network solutions

The previously mentioned planning alternatives are based on system expansion/reinforcement. In contrast, other planning alternatives exploit existing resources in the power system infrastructure without necessarily reinforcing or increasing the system's capacity. These alternatives have been referred by recent literature as non-network solutions [13], having emerged to adapt to changes in energy systems, avoiding unnecessary investments and operational contingencies [102]. Such options are based on using and managing available resources in the power system, e.g., smart energy management, and energy markets.

5.1. Smart energy management

Smart energy management has gained attention in the past few years, mainly in proposals focused on distribution system operation [103,104]. The intelligent coordination of energy generation and consumption is a valuable tool to reduce operational costs, improve distribution system reliability indices, deal with network congestion, manage DERs, and even minimize expansion costs from the grid. Therefore, studies on smart energy management in the DSEP problem are highly required.

With the advancement of smart grids and incentives for energy management strategies, there will be a need to reform investments in distribution networks, encouraging a more intelligent use of current infrastructure; this could mean giving higher returns to projects that utilize more economical solutions, such as efficient DR measures, rather than costly upgrades. Planners must actively seek these cost-effective solutions [105].

Some planning proposals have included smart energy management through the DR approach [86,106]. However, this line of research in planning is still minimal, and more studies are needed to evaluate the impact of intelligent management tools in planning distribution systems.

5.2. Energy markets

The technological revolution, sustainable development, environmental goals, and the increasing penetration of DERs are changing how distribution networks are planned and managed. The new paradigm in distribution systems requires modern solutions for adequate management, operation, and planning. Thus, a new era of the DSEP problem must include non-network planning options (e.g., energy provided by DERs and flexibility services) and the consideration of new agents, such as prosumers. These new configurations allow the emergence of new market designs such as LEM [107] and flexibility markets [108]. However, these markets are often addressed in research from the perspective of operation and management of distribution networks [109], neglecting the crucial impact they will have in the planning phase. In the later section, the consideration of energy markets in DSEP is discussed in detail.

6. Consideration of energy markets in distribution system expansion planning

This section explores the rise of market mechanisms integrated into the DSEP problem by presenting key concepts and reviewing planning strategies considering not only more traditional markets (i.e., wholesale and retail markets) but also new markets such as LEMs, carbon emission trading markets (CET), or flexibility markets.

6.1. Key concepts

This section presents the key concepts involved in the DSEP problem considering market mechanisms. The concepts mentioned in this section were addressed by at least one of the reviewed works and all these concepts will be developed in later sections.

- Wholesale energy market (WEM): The wholesale energy market is a centralized environment for power and electricity trading between participants in the power system. Examples of participants in the WEM are GENCOs (generating companies), transmission companies, DISCOs (distribution companies), large customers and Independent System Operators (ISO) [107,110].
- Retail energy market (REM): The retail energy market allows the final consumer to buy electricity from competitive retail suppliers. Thus, this market becomes more competitive as free retail entry is provided [107].
- Local energy market (LEM): The concept of local energy market has emerged as a mechanism to deal with new actors and technologies in the distribution system. The objective of this market is to encourage small consumers, DER owners, and other stakeholders to sell energy in a competitive local market. In addition, LEM aims to balance supplied energy and demand locally [6,109].
- Flexibility market: The flexibility market brings the opportunity to trade flexibility between the DSO and other participants (e.g., DER owners). Thus, the DSO can procure flexibility services related to congestion management, voltage support, peak shaving, among other services [59].
- Carbon market: This market is designed to save energy and meet environmental goals, collaborating to reduce greenhouse gases [108]. The European Union was the pioneer in creating this market in 2005. It is estimated that since its creation, the emissions in the sectors it covers (e.g., electricity and heat generation sectors, oil refinery sectors) have decreased by 42.8 % [109].

6.2. Proposals on distribution system planning with energy markets

The DSEP is a fundamental aspect of the management of power systems. In traditional distribution networks, the main expansion alternatives are network reconfiguration, construction of new circuits, installation of new transformers/substations, and the allocation of reactive power resources. However, the current integration of DERs and new market concepts that emerged in active distribution systems are significantly changing the DSEP. This section presents a brief summary of the most relevant aspects of recent proposals for solving the planning problem that have addressed these new concepts [41,49,83,94,96,111–123].

Different methods have been applied to optimize the planning of distribution networks and DERs considering energy trading such as mathematical programming [96,111–114,116–119,121,122], computational intelligence [41,115,120], and math-heuristic optimization [49,94,123]. In mathematical programming, the problem has been formulated as a MILP model [111,116,117,122], a MISOCP model [112–114,119], and a MINLP model [96]. The step-controlled primal dual interior point method is used in Ref. [121] to solve the optimal power flow. Computational intelligence methods include GA [120], evolutionary algorithm [41], and back propagation neural network models [115]. Also, methods based on math-heuristic have been used in Refs. [49,94,123], combining the heuristic evolutionary vertical sequencing protocol with the stochastic MILP formulation to solve the optimal Microgrids expansion planning in Ref. [94], two-stage MILP formulation and immune GA in Ref. [123], and GA and MILP formulation in a four-stage optimization model in Ref. [49].

Most of the works reviewed in this section address traditional markets (i.e., WEM, REM) [112,120]. However, not all end-users, especially the residential ones, can participate in these markets due to existing entry barriers [122]. Therefore, LEMs are emerging as an alternative to address this issue, empowering small energy producers and prosumers. The main objective of LEMs is to encourage small consumers, producers, and prosumers to trade power with each other in a competitive market, locally balancing energy supply and demand [124]. In this context, some research has directed efforts to explore the concept of LEMs into the DSEP problem [49,114–117]. For instance, a mathematical-heuristic approach has been proposed in Ref. [49] to solve the coordinated problem of MG expansion planning. In Ref. [114], the authors provide an important contribution to the planning problem by allowing end users to exchange energy directly with each other in a Peer-to-peer (P2P) market (which is a type of LEM). A model for trading-oriented battery energy storage system planning is proposed in Ref. [115], which is formulated as a mutual-iteration and multi-objective two-stage model. The optimization model considers the participation of DERs and storage systems in the decentralized market. The market-clearing mechanism is the averaging pricing market, which is a double-side auction model used in the energy transactions.

Bi-level optimization has often been employed in the planning problem with energy markets [116,117]. In the bi-level planning proposed in Ref. [116], DSO, DER owners, and load aggregators participate in the LEM. Investment decisions on network assets and DERs are defined at the upper level, while the optimal operation of DERs is achieved by clearing the LEM at the lower level. The same authors of [116] proposed a model for the reliability-based DSEP [117] considering LEMs.

In recent years, a significant body of research has focused on addressing the penetration of EV demand in distribution systems and the technologies used to meet this new demand, e.g., gridable parking lots (GPL) [123]. The power injection capability of GPLs is explored as an alternative flexibility resource that can supply power to the network under faulty conditions. Thus, the authors proposed a dynamic bi-level model for the DSEP incorporating the conflicting objectives of the DSO and GPL owners. In a different line of work [111], focused on providing a DSEP strategy coordinated with the bulk power system planning (planning of generation and transmission systems). The authors concluded that optimizing the DSEP problem taking into account the DSO's participation in energy markets can reduce investments in the expansion of bulk power systems and improve electrical system operating rates. Furthermore, in this work, the planning of DERs is carried out by the DSO, which is not an adequate practice in the contemporary landscape [59].

In power systems, there are several types of markets not only limited to the trade of electricity. In fact, there are also other markets, such as distributed capacity markets [118], flexibility markets [83], and CET markets [119]. The DSO manages the network operation and acts as a market supervisor in Ref. [118]. The DISCOs and DER aggregators make bidding plans for the planning of distribution network and DERs and submit them to the DSO. The approach proposed by the authors allows DER aggregators to participate and compete with DISCOs in the distributed capacity market.

Only recently, in 2022, a risk-based DSEP has been proposed considering flexibility markets in Ref. [83]. The article explores flexibility services acquisition as a non-network planning alternative for distribution networks compared to traditional planning options (i.e., substations, circuits, and transformers). The proposed model is developed from the perspective of the DSO, which, to maintain the proper operation of the system, can invest in network reinforcement and obtain flexible services from prosumers. Therefore, the interests of other stakeholders (e.g., DER owners) are not optimized and investments in DERs are disregarded. Another market rarely explored in the DSEP problem is the CET market, created in 2005 by the European Union [125] to support the reduction of CO_2 emissions. Such a market has been considered in the planning strategy proposed in Ref. [119]. In this work, the authors presented a bi-level model to solve the planning problem considering optimizing investments in PV units, WT units, hydraulic turbines, microturbines, and ESSs.

An important aspect to be highlighted is that centralized planning, usually adopted in the DSEP problem, would not work well in proposals that incorporate market transactions, where the interests of different participants must be considered. Finally, the most frequent features in planning proposals that address energy markets are the following: (i) Methods: The methods that work best for this problem are based on Stackelberg games and bi-level programming; (ii) Decentralized planning: The interests of different stakeholders are considered in a decentralized planning strategy; (iii) Problem formulation: The original problem model is often approximated as a MILP or MISOCP model. Such models ensure the solution's optimality.

6.3. *Objective function*

Most of the adopted objective functions aim to minimize the annualized net present value (NPV) of costs in terms of investment (fixed costs) and operation (variable costs). Investments may be related to traditional expansion alternatives and modern expansion alternatives. The former includes investments in substations, transformers, circuits, and reactive power resources, while the latter is related to investments in PV units [94,122], WT units [113,121], dispatchable DG units [111, 115], and ESSs [116,117]. In addition, there are proposals with investments that aim not only to meet the electricity demand but also to meet the heating demand through CHP units and gas-fired boilers [94]. Reliability as an objective function has also been explored in Ref. [120] in which a multi-objective formulation addresses two conflicting objectives: cost minimization and reliability maximization. In addition, bi-level or multi-level formulations [41,94,96,111,113,116,117,123] consider different objective functions for each level. Table 3 presents the objective functions adopted in DSEP problem with energy markets. Such objectives are divided into mono-objective, multi-objective, and multi-level models. It is noted that bi-level or multi-level optimization models are more adapted for proposals incorporating different players' interests (e.g., DSO, DER owners).

6.4. *Constraints*

The most common constraints considered in the planning problem (see Table 4) with market transactions are 1) power balance, 2) operational limits, 3) voltage limits, 4) investment limits, 5) radiality constraints, 6) DER model, 7) reliability constraints, 8) DR constraints, 9) limits of load curtailment, 10) energy transaction constraints, 11) risk measure. Furthermore, additional constraints must be added to the DSEP problem considering new market designs, such as constraints that model the utilization of DERs and the energy transactions in the LEM. Finally, in the next section, the main market characteristics introduced in the DSEP problem are described.

6.5. *Market characteristics*

The reviewed works previously mentioned consider the following types of markets: WEM and/or REM [96,112,120], LEM [114,117,126], capacity market [118], flexibility market [83], and CET market [119]. The WEM and REM are markets targeting large energy producers, large consumers, and retail participants. Thus, due to existing regulatory barriers [127], prosumers and small energy producers cannot participate directly in these markets unless there is an aggregator as intermediary [122]. For example, such aggregation can be done by employing virtual power plants [112,122] and microgrids [94,96,111,122] Furthermore, there is an alternative for prosumers to participate directly in energy trading using LEM [114,126]. The markets in the power system are not limited to electricity trading; there are also markets for other products/services, such as distributed capacity market [118], flexibility market [83], and CET market [119].

Table 5 shows the main aspects of the markets adopted in the planning problem of the reviewed works. Fig. 3 highlights some market configurations applied to the planning proposals reviewed in section 4.3, such as a flexibility market structure (Fig. 3a), a LEM considering a

Table 3
Objective functions of planning proposals with energy markets.

Reference	Objective Function	Method
[119]	Minimization of the investment, operational cost, and carbon emission costs	MISOCP
[120]	Multi-objective: cost minimization and reliability maximization	MINLP
[121]	Maximization of social welfare	Heuristic-based algorithm
[122]	Maximization of social welfare	MILP
[49,114]	Minimization of NPV investment and operational costs	MISOCP
[83]	Minimization of NPV investment and operational costs	LP
	Bi-level or multi-level optimization model	
[41]	Upper-level: Minimization of DSO or DISCO costs Lower-level: Maximization of the interests of DER owners/prosumers	Heuristic-based algorithm
[94]	First level: Minimization of investment and operational costs including reliability Second level: Maximization of the interests of microgrids and DR providers Third level: Minimization of CVaR Fourth level: Minimization of the total interruption costs	GA
[96]	Upper-level: Maximization of the interests of DSO Lower-level: Maximization of the interests of microgrids investors	LP
[123]	Upper-level: Minimization of NPV investment and operational costs including risk measure Lower-level: Maximization of the interests of GPL owners	MILP/IGA
[111]	Upper-level: Minimization of DSO costs Lower-level: Minimization of load curtailments in contingency conditions	MILP
[112]	First level: Minimization of NPV investment and operational costs Second level: Minimization of WT and PV generation curtailments Third level: Minimization of nodal net load variance	MISOCP
[113]	Upper-level: Minimization of DSO or DISCO costs Lower-level: Maximization of the interests of DER owners/prosumers	MISOCP
[115]	Upper-level: Minimization of investment and operating costs including reliability Lower-level: Maximization of social welfare	Levenberg–Marquardt algorithm
[116, 117]	Upper-level: Minimization of DSO or DISCO costs Lower-level: Maximization of the interests of DER owners/prosumers	MILP

Table 4
Most common constraints for the DSEP problem with market transactions.

Reference	Constraints
[41,94,96,111,115,117,119–121,126]	Power balance
[41,94,96,111,115,117,119–121,126]	Operational limits
[41,94,96,111,117,119–121,126]	Voltage limits
[41,117,120,126]	Investment limits
[114,117,120,126]	Radiality constraints
[49,96,111,112,115,117,126]	DER model
[94,111,120]	Reliability constraints
[94,111,114,119,122]	DR model
[94,96]	Limits of load curtailment
[49,94,96,111,112,115,121–123]	Energy transaction constraints
[83,94,111,114,123]	Risk measure

P2P market structure (Fig. 3b), and a WEM and partial REM structure (Fig. 3c). In the configurations shown in Fig. 3a and c, the importance of the DSO is remarkable, acting on several fronts and assuming a variety of responsibilities, e.g., distribution network operation and management, distribution network expansion planning, DERs management, procurement of flexibility services from DERs, and distribution and sale of electricity to the consumers.

The research in DSEP considering new market designs is recent, with many topics yet to be explored in this field; for instance, only one work can be found in the specialized literature regarding DSEP with flexibility markets [83].

Furthermore, LEMs in expansion planning were only started to be addressed in 2020. Many aspects of this market have yet to be explored, such as P2P, community self-consumption, and transactive energy market structures. Therefore, Table 6 summarizes the main aspects of the reviewed works that consider energy markets. Also, this table aims to show the limitations of planning proposals with markets for application in large-scale systems. Furthermore, it is noted that the solution techniques most applied in real systems are based on computational intelligence methods, followed by math-heuristics. Table 7 identifies the gaps found in the planning proposals with energy markets, such gaps were identified from this investigation.

Some works apply their proposals in real systems as in Ref. [119], which addresses the carbon market in the Chinese context; the study [49] cites the European context, more specifically Norway, in a microgrid planning proposal; in Ref. [83] the flexibility market is applied in the Italian context. However, it is essential to highlight that most of the reviewed works operate in a sandbox environment, not considering geographic and regulatory constraints. Different regulations are being developed around the world to address these new markets; thus, implementing them markets will depend on the rules in force in the region where they will be applied.

It is important to point out that new market concepts such as LEMs are still very experimental. While this concept has been explored in research simulations and pilot projects, its large-scale implementation has yet to be a reality, as shown in Table 8. This table highlights relevant developments in the three main electricity markets (European, Chinese, and U.S.), emphasizing that the main developments are still limited and experimental. In this regard, a study published in 2022 [130] identifies the main challenges for the large-scale implementation of these types of markets. Finally, Fig. 4 shows the recent growth of studies on the DSEP problem with energy markets, confirming the upward trend of research in this area.

6.6. DSO responsibilities

The role of selling and buying electricity in the WEM is not a common practice for the DSO; such a procedure is separated from the DSO responsibilities listed in Ref. [137]. However, in some proposals, the DSO assumes responsibility for selling and purchasing electricity [96]. In some cases, the DSO also acts as a supervisor and market facilitator [111, 117]. In addition, there are proposals in which the DSO is responsible for deciding on investments in DERs [94], which might not be appropriate due to conflict of interest with other actors (e.g., DER investors), technical barriers, and existing regulations. If the DSO takes on new responsibilities, appropriate measures must be used to guarantee the DSO does not use access to the data to get a commercial advantage or generate distortions in the energy markets [138]. The DSO's responsibilities include ensuring the operation and maintenance of the distribution system within a specific area, including expansion of the system when necessary. In the new energy scenario, the DSO has been acquiring new responsibilities such as technical validation for energy markets, procurement of grid services from DERs, peak demand control through DERs, network congestion management, or reactive power support for TSOs [59]. The specialized literature shows the evolution of these new DSO responsibilities [83,126]. For example, flexibility

Table 5
Market aspects addressed in planning problems.

Participants	Design								
	1	2	3	4	5	6	7	8	9
DSO	[83,120]	[96]	[115,120]	[117,118]	[111,118]	[114]	[94,112]	[121,123]	–
ISO	[120]	[96]	[120]	–	[111]	–	[94]	–	–
DISCO	[128]	[113]	[120]	[118]	[118]	[114]	–	–	–
GENCO	[120]	–	[120]	–	[111]	–	–	–	–
DER owners	[83]	[113]	[115]	[117,118]	[111]	[114]	[112,116]	[112,121]	[119]
DR providers	–	–	–	–	–	–	[94]	–	–
Aggregators	–	[113]	–	[116,118]	[111]	[118]	[116]	[129]	–
Microgrids	–	[96]	–	–	[111]	–	[94]	–	–
GPL owners	–	–	–	–	–	–	–	[123]	–

1: Bilateral contract; 2: Internal contracts; 3: Two-sided auction model; 4: DLMP signal; 5: LMP signal; 6: Feed-in tariff; 7: Day-ahead market; 8: Real-time market; 9: Green power certificate trading.

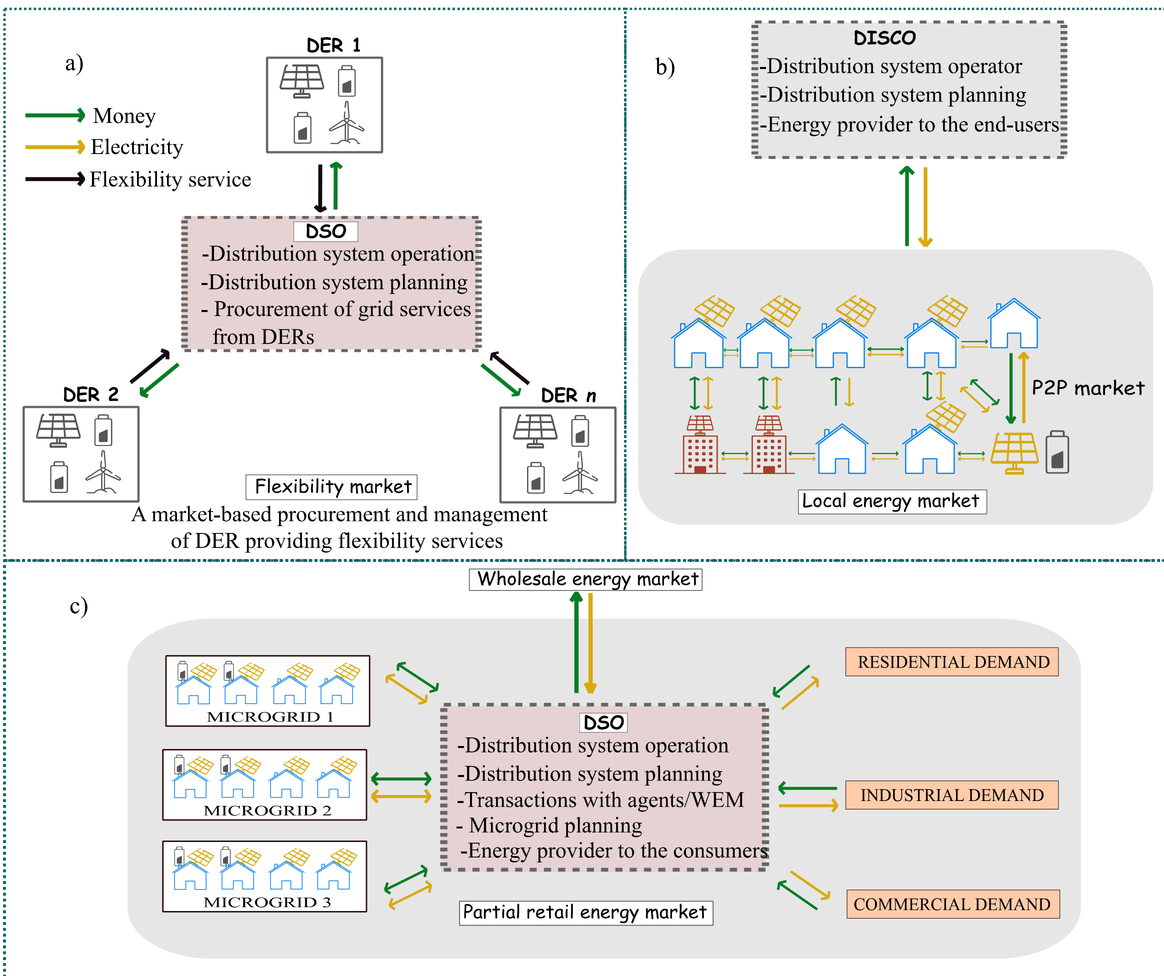


Fig. 3. Market configurations adopted in the DSEP: a) Flexibility market; b) Local market; and c) WEM and REM.

services contract by DSO is considered in Ref. [83]. The local energy market is managed by the DSO in Ref. [126].

For the establishment of these new DSO responsibilities, the development of local markets is a key factor to allow prosumers and DER owners to support the power system by providing flexibility. For this reason, DSEP research should address these new market designs, aiming to contribute to the future of power systems, enabling the distribution system to rapidly and adequately integrate DERs. Finally, the following section discusses and evaluates modern planning alternatives and non-network solutions to the planning problem.

7. Discussion and evaluation

This work addressed modern expansion alternatives and non-network alternatives to the DSEP problem. In such approaches, different types of decision variables, objective functions, constraints, load models, and uncertainty models can be adopted depending on the chosen optimization approach. Decision variables can be classified as follows: installation of new circuits, reinforcement/reconductoring of existing circuits, construction of new substations, reinforcement of existing substations, installation of switching or protection devices, installation of reactive power compensators, allocation of DG units,

Table 6
Taxonomy of DSEP proposals considering energy markets.

Reference	Solution technique	Market Type	Year of Publication	Test system
[41]	2	WEM	2017	33-node
[94]	3	WEM	2018	18 and 33-node
[49]	3	LEM	2019	Real system
[119]	1	CET market	2022	33-node
[113]	1	REM	2020	54-node
[114]	1	LEM	2021	33-node
[115]	2	LEM	2021	24-node
[117]	1	LEM, WEM	2022	54 and 138-node
[118]	1	Capacity market	2021	33-node and real system
[83]	1	Flexibility market	2022	Real system
[120]	2	WEM and REM	2012	Real system
[121]	1	REM	2015	84-node
[122]	1	WEM	2016	–
[96]	1	WEM and REM	2018	33-node
[123]	3	WEM	2018	33 and 118-node
[111]	1	REM and WEM	2019	24-node IEEE RTS
[126]	1	LEM	2021	24 and 86-node
[129]	1	REM	2020	33 and123-node

1: Mathematical programming; 2: computational intelligence methods; 3: math-heuristic optimization

ESSs, and EVCSs. Such decision variables have been frequently addressed in the DSEP in the last decade. In the past few years, new variables have been included in the problem, such as the allocation of hydrogen fueling stations, aiming to contribute to integrating hydrogen fuel vehicles [61]. Furthermore, some proposals not only address the electricity demand but also strive to fulfill the growing need for hydrogen [62], heating [94] and cooling [77]. In this modern planning, in addition to the traditional objective functions related to minimizing investment and operational costs, as well as maximizing reliability, other objectives have emerged, e.g., minimization of greenhouse gases, maximization of the interests of DER owners/prosumers, maximization of the interests of microgrids and DR providers.

To solve the different planning problems, the works reviewed have used the following methods: mathematical programming, heuristics/metaheuristics, hybrid methods. Among the hybrid methods, application of Math-heuristic has emerged; this method combines the advantages of mathematical programming (mainly the guarantee of the exact solution) and heuristics related to its versatility and computational cost, incorporating the "best of both worlds" in a solution method. Other hybrid methods considered in the reviewed works are differential evolution algorithm and GA; differential evolution algorithm/Fuzzy-Pareto Dominance (FPD); and TS/PSO. Fig. 5 illustrates the frequency that each method mentioned is applied in the reviewed works. This figure shows that mathematical programming is the most used method in current planning proposals. In the context of this method, the MILP model is the most used to formulate the problem, with MISOCP formulations being the second most used. Such models ensure the solution's optimality, a valuable aspect of the planning problem. Although these models have a high computational cost, this is not a decisive factor in planning proposals. However, such models may not reach convergence depending on the complexity of the problem. Since modern planning becomes more complex when new technologies and operation standards are adopted, MILP and MISOCP models may not be the most adequate formulation.

Expansion planning is based on increasing the capacity of the distribution network by adding or reinforcing assets in the network to meet the demand at the lowest possible cost in a safe way. Non-network solutions have emerged as an alternative to expansion planning. Such solutions as flexibility services through DERs, smart energy management, and energy markets have the potential to reduce operational costs, deal with network congestion, and use DER more efficiently. With this investigation, it is clear that expansion alternatives are more explored in the planning context [44,139]. However, in the past few years, there has been a tendency in some studies to explore non-network solutions [78,

Table 8
Relevant developments in the three main energy markets.

Market context	Relevant developments
US	P2P energy market applied to a microgrid in Brooklyn, New York [131]
European	Quartierstrom: a P2P market pilot project with 37 residences implemented in Walenstadt, Switzerland [132] GridFlex Heeten: an energy community pilot project in the Netherlands [133] PeerEnergyCloud: a platform for local cloud-based electricity trading in Germany [134] Empower: a trading platform for exchanging energy in local markets [135]
Chinese	List of 26 pilot projects chosen to commercialize energy for the distributed generation market. Such projects have a total installed capacity of 1.47 GW [136]

Table 7
Identified gaps in the specialized literature related to DSEP problem.

Ref.	Flexibility market	LEM	Allocation of EVCS	EV smart charging	Support to TSOs (using LEM)	Energy hub	Long term uncertainty
[41]							
[94]							
[49]		✓					
[119]							
[113]							
[114]		✓					
[115]		✓					
[117]		✓					
[118]							
[83]	✓						
[120]							
[121]							
[122]						✓	
[96]							
[123]			✓	✓			
[111]							
[126]		✓					
[129]							

✓Considered; Empty cell indicates that the reference does not consider the corresponding topic.

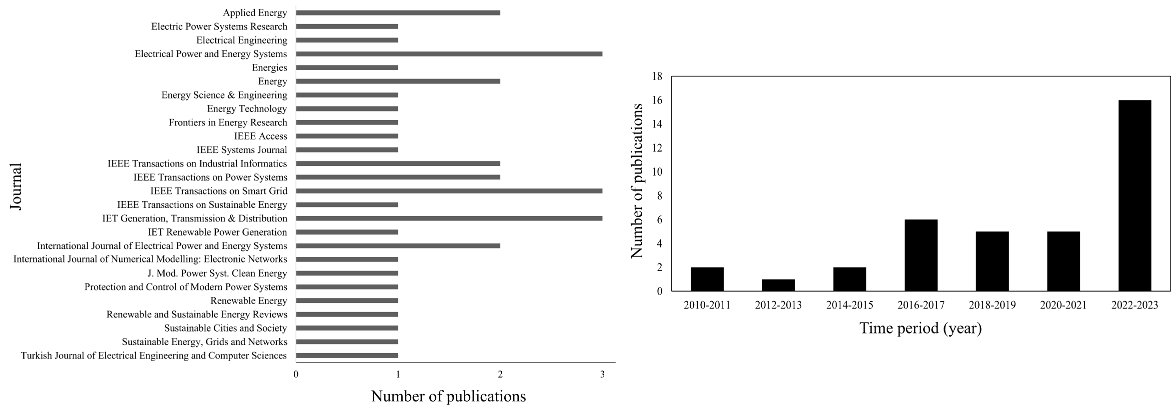
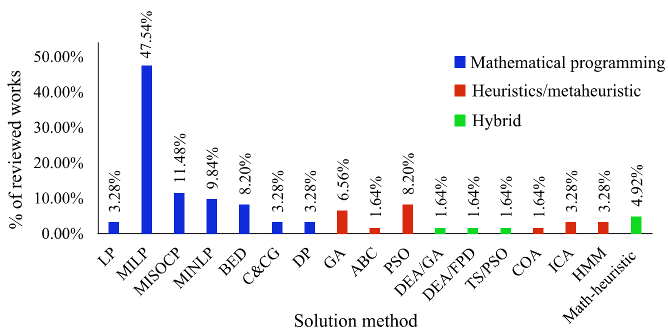


Fig. 4. Number of publications about DSEP problem with energy markets.



LP: Linear programming; BED: Bender's decomposition; C&CG: Column-and-constraint generation;
DP:Dynamic programming; COA: Cuckoo optimization algorithm; ABC: Artificial bee colony;
DEA:Differential evolution algorithm; ICA: Imperialist competitive algoritthm;
HMM: Heuristic moment matching; FPD: Fuzzy-pareto dominance.

Fig. 5. Solution methods for the DSEP problem with flexibility resources.

129]. Such an approach brings the need for integrated optimization of expansion planning and operation in distribution systems since the performance of non-network solutions is present in the system operation stage, which can directly impact network planning decisions. However, only some studies have addressed these alternatives in the context of planning. Thus, the impact of non-network solutions on the DSEP problem should be explored in future proposals.

This research is particularly interested in addressing energy markets in the DSEP problem since the development of anti-monopoly reforms in the energy system in several countries has favored the participation of DERs in market transactions [115]; it is expected that this trend will continue in the coming years with the implementation of new market concepts such as the local market [49,114,126]. Energy markets have been further explored in problems focused on the operation and management of distribution systems [140]. Unfortunately, most planning research in the specialized literature does not consider energy trading in their formulations [69,80,85]. However, a planning approach that does not address energy markets might not be the most appropriate since such markets impact the DSEP and can contribute to avoiding or postponing investments in distribution network expansion [83]. Therefore, planning strategies that ignore these elements can lead to overinvestment and oversized distribution networks.

Table 9 presents the studies reviewed in this work in chronological order, highlighting the considered planning alternatives (traditional, modern, non-network solutions). Traditional alternatives to the DSEP problem focus on expanding the physical infrastructure, considering investments in substations, circuits, capacitor banks, and voltage regulators. In contrast, modern alternatives integrate DERs (e.g., PV units, WT units, and ESS). In modern expansion alternatives, decision variables involve allocating and sizing DER technologies to meet demand and reinforce the distribution system. Unlike traditional/modern expansion alternatives, non-network solutions do not increase system capacity by reinforcing or expanding the network or adding new assets. Non-network solutions exploit resources within the power system infrastructure without physical expansion. An example of non-network solutions is energy management strategies such as DR. Significantly, there has been a growing interest over time in exploring non-network solutions within the DSEP problem. Another relevant point presented in this table is that, since 2016, research has emerged in the area considering different energy resources in addition to electricity, such as hydrogen and heating.

The investigation of the state of the art for the DSEP problem shows that the current planning proposals are partially aligned with the changes that have been taking place in the power system, different from the proposals focused on the operation of distribution systems. It is interesting to note that proposals focused on the operation have been more committed to addressing new actors in the power system, such as flexibility services, smart energy management, and energy trading.

Table 10 presents the three main approaches most found in the revised planning proposals, highlighting their strengths and weaknesses. Such approaches, strategies, and methods must be adapted to modern distribution systems. Centralized planning strategies based on unilateral decisions are not the most appropriate. Thus, the interests of other actors, such as prosumers and EVCS investors, must be incorporated. Planning alternatives based solely on system expansion can lead to overinvestments, so non-network solutions such as smart energy management should be explored in conjunction with network expansion. Therefore, these new operating standards and technologies make the planning problem even more complex. As a result, the most used solution method for this problem, mathematical programming (see Fig. 5), may not show convergence in modern planning problems. In this context, exploration of new methods is highly required to address the challenges of the new era planning problem. Finally, Fig. 6 summarizes the main points of this research, highlighting the research question, findings, and future directions.

8. Trends and future research directions

This investigation of the current literature shows a trend in the planning of distribution systems to address non-network solutions, which increases the complexity of the problem by promoting the integrated optimization of planning and operation in the distribution network. Therefore, the new era of the DSEP problem calls for developing new strategies to adequately address the technologies and improved operation configurations in the power system. Thus, we highlight the main trends and future research directions for the planning of modern distribution systems:

Table 9
Planning alternatives and non-network solutions addressed for the DSEP problem.

Year of Publication	Ref.	Expansion alternatives		Non-network solutions		Type of demand
		Traditional	Modern	Smart energy management	Energy market	
2010	[141]	x	✓	x	✓	electricity
2011	[142]	✓	✓	x	✓	electricity
2012	[120]	✓	x	x	✓	electricity
2012	[18]	✓	x	x	x	electricity
2012	[143]	✓	x	x	x	electricity
2012	[20]	✓	x	x	x	electricity
2012	[21]	✓	x	x	x	electricity
2012	[72]	x	✓	x	x	electricity
2013	[54]	x	✓	x	x	electricity
2013	[144]	✓	x	x	x	electricity
2013	[145]	✓	x	x	x	electricity
2013	[146]	✓	x	x	x	electricity
2013	[38]	✓	x	x	x	electricity
2014	[31]	✓	x	x	x	electricity
2014	[39]	✓	x	x	x	electricity
2014	[147]	✓	✓	✓	x	electricity
2015	[26]	✓	x	x	x	electricity
2015	[37]	✓	x	x	x	electricity
2015	[50]	x	✓	x	x	electricity
2015	[148]	✓	✓	x	x	electricity
2015	[121]	x	✓	x	✓	electricity
2016	[24]	✓	x	x	x	electricity
2016	[75]	x	✓	x	x	electricity, heating
2016	[149]	✓	x	x	x	electricity
2016	[89]	x	✓	x	x	electricity
2016	[106]	✓	✓	✓	x	electricity
2016	[81]	x	✓	x	x	electricity
2016	[56]	✓	x	✓	x	electricity
2016	[150]	x	✓	✓	✓	electricity
2016	[122]	x	✓	x	✓	electricity
2017	[48]	✓	✓	x	x	electricity
2017	[151]	✓	✓	x	x	electricity
2017	[55]	✓	✓	x	x	electricity
2017	[64]	✓	✓	x	x	electricity, heating
2017	[152]	x	✓	x	✓	electricity, heating
2017	[153]	x	✓	x	✓	electricity
2017	[41]	✓	✓	x	✓	electricity
2017	[154]	x	✓	x	✓	electricity
2018	[15]	✓	✓	x	x	electricity
2018	[155]	✓	x	x	x	electricity
2018	[27]	✓	x	x	x	electricity
2018	[43]	✓	x	x	x	electricity
2018	[76]	✓	✓	x	x	electricity
2018	[42]	✓	✓	x	x	electricity
2018	[78]	✓	✓	✓	x	electricity
2018	[57]	x	✓	✓	x	electricity
2018	[58]	✓	✓	x	x	electricity
2018	[88]	x	✓	x	x	electricity
2018	[94]	✓	✓	✓	✓	electricity
2018	[96]	✓	✓	x	✓	electricity
2018	[111]	✓	✓	✓	✓	electricity
2019	[156]	x	✓	✓	✓	electricity
2019	[98]	x	✓	x	x	electricity
2019	[9]	✓	✓	x	x	electricity
2019	[80]	✓	✓	✓	x	electricity
2019	[85]	✓	✓	x	x	electricity
2019	[30]	✓	x	x	x	electricity
2019	[77]	✓	✓	✓	x	electricity, heating
2019	[157]	x	✓	x	x	electricity
2019	[40]	✓	✓	x	x	electricity
2019	[70]	✓	✓	x	x	electricity
2019	[158]	✓	✓	x	x	electricity
2019	[49]	x	✓	x	✓	electricity
2020	[28]	✓	x	x	x	electricity
2020	[159]	x	✓	x	x	electricity
2020	[87]	✓	✓	x	x	electricity
2020	[62]	x	✓	x	x	electricity, heating, hydrogen
2020	[51]	✓	✓	x	x	electricity
2020	[160]	✓	✓	x	x	electricity, heating
2020	[66]	✓	✓	x	x	electricity
2020	[161]	x	✓	✓	✓	electricity
2020	[113]	✓	✓	✓	✓	electricity
2020	[129]	✓	✓	✓	✓	electricity

(*continued on next page*)

Table 9 (*continued*)

Year of Publication	Ref.	Expansion alternatives		Non-network solutions		Type of demand
		Traditional	Modern	Smart energy management	Energy market	
2021	[16]	✓	✓	x	x	electricity
2021	[25]	✓	x	x	x	electricity
2021	[162]	✓	✓	x	x	electricity
2021	[139]	✓	✓	x	x	electricity
2021	[163]	✓	✓	x	x	electricity
2021	[164]	✓	x	✓	x	electricity
2021	[115]	x	✓	x	✓	electricity
2021	[165]	✓	✓	✓	✓	electricity, heating
2021	[118]	✓	✓	x	✓	electricity
2022	[8]	✓	✓	x	x	electricity
2022	[29]	✓	x	x	x	electricity
2022	[166]	x	✓	x	x	electricity
2022	[61]	✓	✓	x	x	electricity, hydrogen
2022	[65]	✓	✓	x	x	electricity, heating
2022	[167]	x	✓	x	x	electricity
2022	[168]	x	✓	x	✓	electricity, heating
2022	[169]	x	✓	✓	✓	electricity
2022	[97]	x	✓	x	✓	electricity
2022	[170]	x	✓	x	✓	electricity, heating
2022	[83]	✓	✓	x	✓	electricity
2022	[126]	✓	✓	x	✓	electricity
2022	[119]	✓	✓	x	✓	electricity
2022	[171]	✓	x	x	✓	electricity
2022	[117]	✓	✓	✓	✓	electricity
2023	[53]	✓	✓	x	x	electricity
2023	[172]	x	✓	x	✓	electricity
2023	[173]	x	✓	✓	✓	electricity, heating
2023	[174]	✓	x	x	✓	electricity
2023	[175]	✓	✓	✓	✓	electricity
2023	[176]	✓	✓	✓	✓	electricity
2023	[177]	✓	✓	✓	✓	electricity
2023	[178]	✓	✓	x	✓	electricity

✓ Considered; x: Not considered.

Table 10
Main characteristics of the reviewed works.

Approach		Strengths	Weaknesses
Planning strategy	Centralized	Easy implementation, simplification of the problem	Decisions based on the interests of a single actor, usually DSO or DISCOs
Planning alternatives	Traditional and modern investments	System updated according to expected demand	Decisions based solely on expansion may realize over-investments
Method	MILP model	Easy implementation, guarantee of optimality of the solution	May present convergence problems according to the complexity of the problem

Expansion of DSO responsibilities: DSO has an important role in planning; in the context of the current literature, the DSO is traditionally responsible for expanding and operating the distribution network. In addition, in some proposals, the DSO assumes other responsibilities, such as defining investments related to DG units and purchasing and selling electricity. Such practices are not common to the DSO and they are not allowed in some localities due to the conflict of interests of the different stakeholders [59]. Thus, future proposals must pay attention to the boundaries of DSO operations. In addition, the investigation shows a trend towards expanding the DSO's responsibilities, an issue that should be addressed in future planning proposals.

Local market: DSEP research should address new market designs, aiming to contribute to the future of power systems, enabling the distribution system for the rapid and adequate integration of DERs. The few works that address the local market in the planning problem show the impact of this market on expansion decisions, contributing to the reduction of investments in the distribution network. However, the impact of this market on the DSEP problem needs to be further studied to assess the consequences of energy trading among end-users on the expansion of the distribution network. Furthermore, LEMs in expansion planning were only addressed three years ago. Therefore, many aspects of this market have yet to be explored, such as P2P, community self-consumption, and transactive energy market structures. Finally, future research should investigate the local market's potential as a non-network alternative in planning proposals. In this regard, an integration between different non-network solutions, such as the local market and smart energy management, may be carried out in future investigations, highlighting the impact of these solutions on the DSEP problem.

Flexibility market: Based on this study, an effort can be seen in the planning proposals towards non-network solutions to support the integration of DERs and postpone or avoid investments in the distribution network (e.g., investment in line construction). In this context, one of the reviewed works proposed flexibility markets in the planning problem [83]. However, this market is still developing, and its impact on the DSEP problem has not been evaluated. Thus, the new planning proposals should also consider flexibility procurement. Flexibility markets can provide services to the distribution system through DERs, offering the opportunity for end users to participate in these markets. Such markets may be technically regulated and validated by the DSO in conjunction with the TSO [59].

Integrated planning of transmission and distribution systems: A recent research has proposed the integrated planning of distribution and transmission systems [162], aiming at the lowest possible expected cost

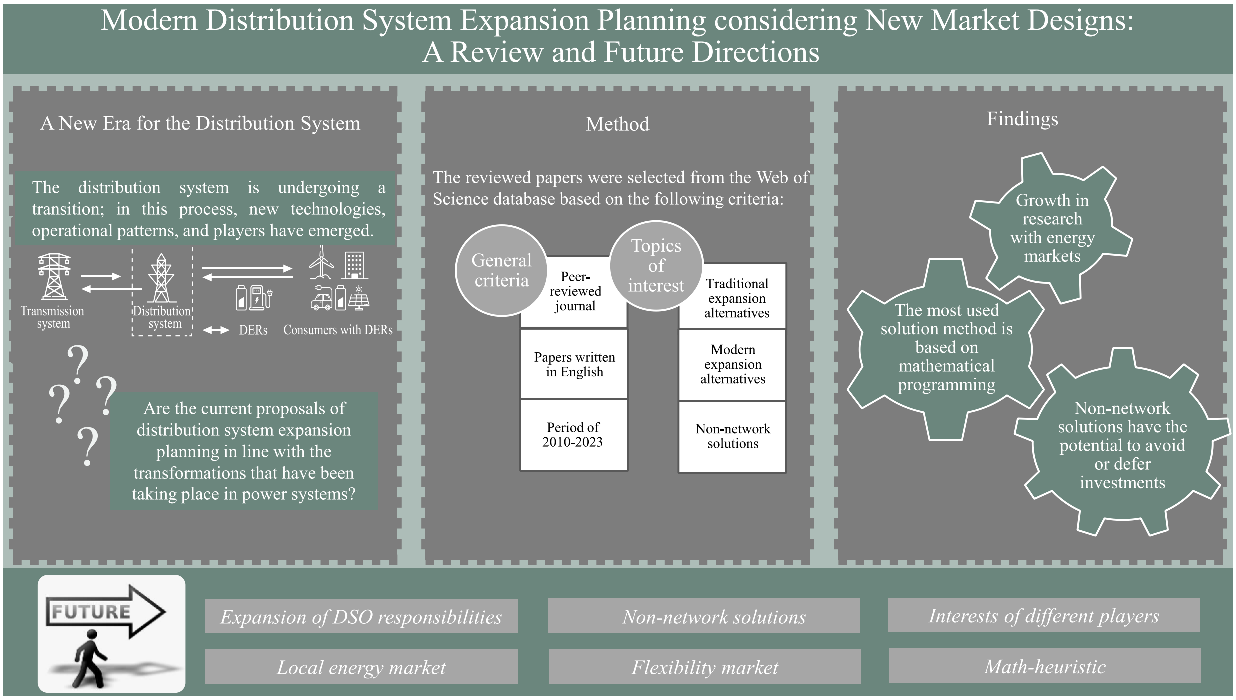


Fig. 6. Overview of this work.

for the entire system (i.e., transmission and distribution systems). However, research in this area is very limited, and the value of integrating these two plans needs to be appropriately evaluated; moreover, coordinated planning of T&D networks and DERs under market environment has not been studied in existing research. Therefore, the modern DSEP must propose a market-based coordination in transmission and distribution network planning.

Interests of different players: Most of the proposed works in the area used a centralized approach to consider investments in the distribution network. However, this practice is not adequate due to conflict of interests with other stakeholders (e.g., DER investors), technical barriers, and existing regulations. Thus, the proposals for the modern DSEP must include the interest and objectives of different players in the decision of investments in the distribution network.

Integration of different energy resources: Some authors have defended the idea of promoting the DSEP problem integrated into the planning of other energy resources (e.g. heating systems, hydrogen, and natural gas) [64,65]. In this context, a traditional distribution system planning with investments in substations and circuits has been addressed in conjunction with gas network planning [65]. The integration of such resources can contribute to improving energy consumption and generation efficiency [62]. In this context, the concept of EH have been explored in a few works [63] being a promising opportunity for future research.

Math-heuristic: As discussed in section 5, the most used method to solve the planning problem is mathematical programming, which mainly uses a MILP model to formulate the problem. Such a formulation may present convergence difficulties as the complexity of the problem increases. Thus, new methods must be developed for future distribution networks, such as hybrid methods based on metaheuristics and mathematical programming, i.e., math-heuristic technique [49]. The use of meta-heuristics offers an advantage in terms of computational time by compromising the exact optimal solution. In other ways, mathematical programming assures an exact solution but presents issues dealing with large-scale complex problems. Furthermore, problems with many variables and operation scenarios may not converge in mathematical programming methods. Thus, combining meta-heuristics and mathematical programming might result in a "best of both worlds" solution. Some DSEP proposals [49,123], considering markets, have shown a trend to use math-heuristic techniques to solve the planning problem.

9. Conclusions

The distribution system is going through significant transformations related to decarbonization, the rapid integration of renewable generation, the emergence of new technologies, new operating standards, and new players. Thus, the distribution system is transitioning to a more decentralized and complex system with the integration of renewable generation and the active participation of end users. In this context, the distribution system expansion planning (DSEP) plays a crucial role in meeting this new configuration, ensuring adequate operation performance regarding reliability, safety, power quality, environmental goals, and sustainability. All these aspects are driving the DSEP problem into a new era. Thus, this work critically presents a critical review of DSEP proposals published in the last decade that consider flexibility resources, energy markets, microgrids, and EV (electric vehicle) charging stations.

The main guideline of this research was based on the following research question: Are the current proposals of distribution system expansion planning in line with the transformations that have been taking place in power systems? From the literature investigation, the conclusion is that the current proposals are partially aligned with these changes. The reviewed works have focused on considering distributed energy resources (DERs) as an expansion alternative in the planning problem. However, the exploitation of flexibility services through these resources must be properly addressed. The investigated proposals mainly address the planning of distribution systems by increasing the system's capacity through the reinforcement or installation of assets. Conversely, some works indicate a tendency to use non-network solutions in the DSEP problem. However, such studies are still limited, and the impact of smart energy management and energy markets on traditional DSEP has yet to be investigated. Therefore, new solution methods are required to explore these aspects in future investigations.

The centralized strategy has often been adopted in the planning of EV charging stations, distributed generation units, and distribution systems. In such approach, planning decisions are made simultaneously based on the interests of a single agent. Furthermore, in some proposals, the distribution system operator (DSO) or distribution companies are responsible for planning the distribution network and DERs, being the very owners in some cases. However, this practice may need to be revised since the existing legislation does not allow the DSO to own and operate the DERs. In this context, planning proposals that address the interests of multiple actors are required. Another critical point to highlight is the integrated planning of multiple energy systems (e.g., electricity, heating systems, hydrogen, and natural gas). Some surveyed works show a need for research in the area of DSEP that addresses this topic.

This research reveals that mathematical programming is the most used method in the current literature. In this context, the problems are mainly formulated as mixed-integer linear programming or mixed-integer second-order cone programming models, guaranteeing the solution's optimality. Nevertheless, they can present convergence problems according to the complexity of the problem. Therefore, new planning proposals must adapt existing methods or develop new methods to solve the modern planning problem; for this purpose, math-heuristic based methods seem promising. Finally, future research on DSEP may include interactions between transmission and distribution systems, local market, flexibility market, and the coordination and consideration of objectives and interests from different (and even some new) actors participating in the system. Furthermore, new methods should be explored to solve the planning problem (e.g., math-heuristic).

This analysis and review of the DSEP problem serve as a valuable tool and guide for stakeholders, offering insights into current practices and emerging trends in distribution system expansion planning. It establishes a basis for future investigations in the area. Furthermore, for the industry, this work highlights the need to review investment strategies to prioritize adopting flexible and renewable energy solutions, thus promoting sustainability, and capitalizing on opportunities in the energy market.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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